

The nycbullets package

New York City Subway route bullets

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1 Introduction

This package typesets the route “bullets” of the New York City Subway:   ,  , , and so on. The artwork comes from the [New York City Subway bullets category on Wikimedia Commons](#), where nearly all of it is in the public domain (see section 3 for the exceptions). It has been compiled into a family of OpenType color fonts, one per set of bullets on Commons, plus one font containing all of them.

2 Usage

Load the package with Lua^LTeX:

```
\usepackage{nycbullets}
```

`\nycbullet` `\nycbullet[options]{key}`

Typesets the bullet named *key* from the current set. The *options* are any of the options of `\nycbulletsetup`, applied to this bullet only; as a shortcut, an option that is just the name of a set selects that set, so `\nycbullet[helvetica]{F}` gives the Helvetica-set .

A *key* names the route, followed by modifiers and variants:

- The route itself: F, 7, SIR, JFK, 10, or – for the dash bullet.
- A lower-case d for the diamond (express) form: Fd , 6d . A few R32 “special” forms add an s: Bs, Bsd.
- Variants, introduced by a dot: colors such as M.brown , historical routings such as B.broadway, years such as AA.1967–1979, and so on.

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The complete list of keys in each set is in section 4. An unknown key is an error.

`\nycbulletsetup` `\nycbulletsetup{⟨key=value list⟩}`

Sets options for all following bullets in the current group. The same options can be given when loading the package.

set=⟨name⟩ The bullet set to use. The sets are `std` (or `standard`), `helv` (helvetica), `nycta`, `r62a`, `r32`, `retired`, `legacy`, and `all`. The default is `std`.

scale=⟨number⟩ Size of the bullets relative to the current font size. The default, 1, makes a circular bullet 0.8 em across, a little taller than a capital letter.

raise=⟨dimension⟩ Shift the bullets up (or down, if negative). The default is 0pt: circular bullets then extend 0.1 em below the baseline.

color=true|false Whether to use the colored glyphs (the default) or the monochrome outlines.

renderer=node|harfbuzz Which luaotfload shaper to use. Both render the bullets in color.

The `color` and `renderer` options take effect when a set's font is first used, so set them in the preamble.

`\nycbulletfont` `\nycbulletfont[⟨set⟩]`

Switches to the bullet font of a set, for typing keys directly. The keys are ligatures in the font, so `{\nycbulletfont F Fd SIR}` gives   . Characters that are not part of a key come out as empty boxes. Separate adjacent bullets with spaces: “AC” is not the same as “A C”.

2.1 Engines

The bullets are in color only with `Lua $\LaTeX$` , which renders the fonts' COLR table. With `Xe $\LaTeX$` the package works, but the bullets come out in a single color (the current text color), because `Xe $\TeX$` does not support color fonts. `pdf $\LaTeX$` cannot use the fonts at all.

2.2 Using the fonts elsewhere

The fonts `NYCSubwayBullets-*.ttf` can be installed and used in any application. Besides the COLR table they have an SVG color table (for Safari, macOS and Adobe applications) and monochrome outlines (as a fallback). Each bullet can be typed as its key, provided the application applies standard ligatures, or as its Private Use Area code point, which is listed in the tables below.

3 Credits

The bullet artwork comes from Wikimedia Commons. Nearly all of it is the work of the Metropolitan Transportation Authority and is in the public domain. These bullets are used under licenses that require attribution:

- [Special \(1967-1979\)](#) by Avy9ishere, CC BY 4.0.

4 The bullets

4.1 NYC Subway Bullets Standard (set=std)

Bullet	Key	Code point	Description
	–	U+E000	Blank
	–d	U+E001	Blank
	0	U+E002	42nd Street Shuttle
	0d	U+E003	Unassigned
	1	U+E004	Broadway–Seventh Avenue Local
	1d	U+E005	Broadway–Seventh Avenue Express
	2	U+E006	Seventh Avenue Express
	2d	U+E007	Seventh Avenue Express
	3	U+E008	Seventh Avenue Express
	3d	U+E009	Seventh Avenue Express
	4	U+E00A	Lexington Avenue Express
	4d	U+E00B	Lexington Avenue Express
	5	U+E00C	Lexington Avenue Express
	5d	U+E00D	Lexington Avenue Express
	6	U+E00E	Lexington Avenue Local
	6d	U+E00F	Pelham Express
	7	U+E010	Flushing Local
	7d	U+E011	Flushing Express
	8	U+E012	Unassigned
	8.turquoise	U+E013	Third Avenue Local
	8d	U+E014	Unassigned
	8d.turquoise	U+E015	Third Avenue Express
	9	U+E016	Broadway–Seventh Avenue Local
	9d	U+E017	Broadway–Seventh Avenue Express
	10	U+E018	Unassigned
	11	U+E019	Unassigned
	12	U+E01A	Unassigned
	13	U+E01B	Unassigned
	A	U+E01C	Eighth Avenue Express

Bullet	Key	Code point	Description
	A.gray	U+E01D	Rockaway Park Shuttle
	Ad	U+E01E	Eighth Avenue Express
	Ad.gray	U+E01F	Rockaway Park Shuttle Express
	B	U+E020	Sixth Avenue Express
	Bd	U+E021	Sixth Avenue Express
	Bs	U+E022	West End Local via Broadway (1986–1988)
	Bsd	U+E023	West End Express via Broadway (1986–1988)
	C	U+E024	Eighth Avenue Local
	Cd	U+E025	Eighth Avenue Express
	D	U+E026	Sixth Avenue Express
	Dd	U+E027	Sixth Avenue Express
	Ds	U+E028	Brighton Local via Broadway (1986–1988)
	Dsd	U+E029	Brighton Express via Broadway (1986–1988)
	E	U+E02A	Eighth Avenue Local
	Ed	U+E02B	Eighth Avenue Express
	F	U+E02C	Sixth Avenue Local
	Fd	U+E02D	Culver Express
	G	U+E02E	Brooklyn–Queens Crosstown Local
	Gd	U+E02F	Brooklyn–Queens Crosstown Express
	H	U+E030	Rockaway Park Shuttle
	Hd	U+E031	Rockaway Park Shuttle Express
	I	U+E032	Unassigned
	Id	U+E033	Unassigned
	J	U+E034	Nassau Street Local
	Jd	U+E035	Nassau Street Express
	K	U+E036	Eighth Avenue Local
	Kd	U+E037	Eighth Avenue Express
	L	U+E038	14th Street–Canarsie Local
	Ld	U+E039	14th Street–Canarsie Express
	M	U+E03A	Sixth Avenue Local
	M.brown	U+E03B	Nassau Street Local (1979–2010)
	Md	U+E03C	Sixth Avenue Express

Bullet	Key	Code point	Description
	Md.brown	U+E03D	Nassau Street Express (2001–2010)
	N	U+E03E	Broadway Express
	Nd	U+E03F	Broadway Express
	O	U+E040	Unassigned
	Od	U+E041	Unassigned
	P	U+E042	Unassigned
	Pd	U+E043	Unassigned
	Q	U+E044	Second Avenue Express
	Q.orange	U+E045	Sixth Avenue Express (1988–2001)
	Qd	U+E046	Second Avenue Express
	Qd.orange	U+E047	Sixth Avenue Express (1988–2001)
	R	U+E048	Broadway Local
	R.brown	U+E049	Nassau Street Local (1985–1987)
	Rd	U+E04A	Broadway Express
	Rd.brown	U+E04B	Nassau Street Express (1985–1987)
	S	U+E04C	Shuttle
	Sd	U+E04D	Shuttle Express
	S6	U+E04E	Sixth Avenue Shuttle
	S6d	U+E04F	Sixth Avenue Shuttle Express
	SB	U+E050	63rd Street Shuttle
	SBd	U+E051	63rd Street Shuttle Express
	SF	U+E052	Franklin Avenue Shuttle
	SF.silver	U+E053	Franklin Avenue Shuttle
	SFd.silver	U+E054	Franklin Avenue Shuttle Express
	SIR	U+E055	Staten Island Railway
	SIRd	U+E056	Staten Island Railway Express
	SR	U+E057	Rockaway Park Shuttle
	SR.blue	U+E058	Rockaway Park Shuttle
	SRd.blue	U+E059	Rockaway Park Shuttle Express
	T	U+E05A	Second Avenue Local (proposed)
	T.black	U+E05B	Unassigned
	Td	U+E05C	Second Avenue Express (proposed)

Bullet	Key	Code point	Description
	Td.black	U+E05D	Unassigned
	U	U+E05E	Unassigned
	Ud	U+E05F	Unassigned
	V	U+E060	Sixth Avenue Local
	Vd	U+E061	Sixth Avenue Express
	W	U+E062	Broadway Local
	Wd	U+E063	Broadway Express
	X	U+E064	Placeholder for proposed services
	Xd	U+E065	Placeholder for proposed services
	Y	U+E066	Unassigned
	Yd	U+E067	Unassigned
	Z	U+E068	Nassau Street Express
	Zd	U+E069	Nassau Street Express

4.2 NYC Subway Bullets Helvetica (set=helv)

Bullet	Key	Code point	Description
	1	U+E000	Broadway–Seventh Avenue Local
	2	U+E001	Seventh Avenue Express
	3	U+E002	Seventh Avenue Express
	4	U+E003	Lexington Avenue Express
	4d	U+E004	Lexington Avenue Express
	5	U+E005	Lexington Avenue Express
	5d	U+E006	Lexington Avenue Express
	6	U+E007	Lexington Avenue Local
	6d	U+E008	Pelham Express
	7	U+E009	Flushing Local
	7d	U+E00A	Flushing Express
	9	U+E00B	Broadway–Seventh Avenue Local
	A	U+E00C	Eighth Avenue Express
	A.gray	U+E00D	Rockaway Park Shuttle

Bullet	Key	Code point	Description
	A. orange	U+E00E	Eighth Avenue Express via Sixth Avenue
	Ad	U+E00F	Eighth Avenue Express
	B	U+E010	Sixth Avenue Express
	B. yellow	U+E011	Broadway Express (1986–1988)
	Bd	U+E012	Sixth Avenue Express
	Bd. yellow	U+E013	Broadway Express (1986–1988)
	C	U+E014	Eighth Avenue Local
	Cd	U+E015	Eighth Avenue Express
	D	U+E016	Sixth Avenue Express
	D. yellow	U+E017	Sixth Avenue Express, Bronx–34th Street (1986–1988)
	Dd	U+E018	Sixth Avenue Express
	E	U+E019	Eighth Avenue Local
	F	U+E01A	Sixth Avenue Local
	Fd	U+E01B	Culver Express
	G	U+E01C	Brooklyn–Queens Crosstown Local
	H	U+E01D	Rockaway Park Shuttle
	J	U+E01E	Nassau Street Local
	Jd	U+E01F	Nassau Street Express
	K	U+E020	Eighth Avenue Local
	K. gray	U+E021	Canarsie skip-stop (proposed)
	L	U+E022	14th Street–Canarsie Local
	M	U+E023	Sixth Avenue Local
	M. brown	U+E024	Nassau Street Local (1979–2010)
	Md. brown	U+E025	Nassau Street Express (2001–2010)
	N	U+E026	Broadway Express
	Nd	U+E027	Broadway Express
	Q	U+E028	Second Avenue Express
	Qd	U+E029	Second Avenue Express
	R	U+E02A	Broadway Local
	Rd	U+E02B	Broadway Express
	S	U+E02C	Shuttle
	S. blue	U+E02D	Rockaway Park Shuttle

Bullet	Key	Code point	Description
	S.orange	U+E02E	63rd Street Shuttle
	S.yellow	U+E02F	Lefferts Boulevard Shuttle
	SF	U+E030	Franklin Avenue Shuttle
	SIR	U+E031	Staten Island Railway
	SR	U+E032	Rockaway Park Shuttle
	T	U+E033	Second Avenue Local (proposed)
	V	U+E034	Sixth Avenue Local
	W	U+E035	Broadway Local
	Wd	U+E036	Broadway Express
	Z	U+E037	Nassau Street Express
	Zd	U+E038	Nassau Street Express

4.3 NYC Subway Bullets NYCTA (set=nycta)

Bullet	Key	Code point	Description
	1	U+E000	Broadway–Seventh Avenue Local
	2	U+E001	Seventh Avenue Express
	3	U+E002	Seventh Avenue Express
	4	U+E003	Lexington Avenue Express
	4d	U+E004	Lexington Avenue Express
	5	U+E005	Lexington Avenue Express
	5d	U+E006	Lexington Avenue Express
	6	U+E007	Lexington Avenue Local
	6d	U+E008	Pelham Express
	7	U+E009	Flushing Local
	7d	U+E00A	Flushing Express
	9	U+E00B	Broadway–Seventh Avenue Local
	A	U+E00C	Eighth Avenue Express
	A.sixth	U+E00D	Eighth Avenue Express via Sixth Avenue
	B	U+E00E	Sixth Avenue Express
	B.broadway	U+E00F	Broadway Express

Bullet	Key	Code point	Description
	Bd	U+E010	Sixth Avenue Express
	Bd.broadway	U+E011	Broadway Express
	C	U+E012	Eighth Avenue Local
	Cd	U+E013	Eighth Avenue Express
	D	U+E014	Sixth Avenue Express
	D.broadway	U+E015	Broadway Express
	E	U+E016	Eighth Avenue Local
	F	U+E017	Sixth Avenue Local
	Fd	U+E018	Culver Express
	G	U+E019	Brooklyn–Queens Crosstown Local
	H	U+E01A	Rockaway Park Shuttle
	J	U+E01B	Nassau Street Local
	Jd	U+E01C	Nassau Street Express
	K	U+E01D	Eighth Avenue Local
	K.canarsie	U+E01E	Canarsie skip-stop (proposed)
	L	U+E01F	14th Street–Canarsie Local
	M	U+E020	Sixth Avenue Local
	M.nassau	U+E021	Nassau Street Local
	Md.nassau	U+E022	Nassau Street Express
	N	U+E023	Broadway Express
	Nd	U+E024	Broadway Express
	Q	U+E025	Second Avenue Express
	Q.sixth	U+E026	Sixth Avenue Express (1988–2001)
	Qd	U+E027	Second Avenue Express
	R	U+E028	Broadway Local
	Rd	U+E029	Broadway Express
	Rd.nassau	U+E02A	Nassau Street Express
	S	U+E02B	Shuttle
	S.63	U+E02C	63rd Street Shuttle
	S.rockaway	U+E02D	Rockaway Park Shuttle
	S.sixth	U+E02E	Sixth Avenue Shuttle
	T	U+E02F	Second Avenue Local (proposed)

Bullet	Key	Code point	Description
	V	U+E030	Sixth Avenue Local
	W	U+E031	Broadway Local
	Wd	U+E032	Broadway Express
	Z	U+E033	Nassau Street Express

4.4 NYC Subway Bullets R62A (set=r62a)

Bullet	Key	Code point	Description
	–	U+E000	Blank
	1	U+E001	Broadway–Seventh Avenue Local
	2	U+E002	Seventh Avenue Express
	3	U+E003	Seventh Avenue Express
	4	U+E004	Lexington Avenue Express
	5	U+E005	Lexington Avenue Express
	5d	U+E006	Lexington Avenue Express
	6	U+E007	Lexington Avenue Local
	6d	U+E008	Pelham Express
	7	U+E009	Flushing Local
	7d	U+E00A	Flushing Express
	8	U+E00B	Unassigned
	9	U+E00C	Broadway–Seventh Avenue Local
	10	U+E00D	Unassigned
	11	U+E00E	Unassigned
	12	U+E00F	Unassigned
	13	U+E010	Unassigned
	S	U+E011	Shuttle

4.5 NYC Subway Bullets R32 (set=r32)

Bullet	Key	Code point	Description
	A	U+E000	Eighth Avenue Express
	B	U+E001	Sixth Avenue Express
	Bd	U+E002	Sixth Avenue Express
	Bs	U+E003	West End Local via Broadway (1986–1988)
	Bsd	U+E004	West End Express via Broadway (1986–1988)
	C	U+E005	Eighth Avenue Local
	Cd	U+E006	Eighth Avenue Express
	D	U+E007	Sixth Avenue Express
	Ds	U+E008	Brighton Local via Broadway (1986–1988)
	E	U+E009	Eighth Avenue Local
	F	U+E00A	Sixth Avenue Local
	G	U+E00B	Brooklyn–Queens Crosstown Local
	H	U+E00C	Rockaway Park Shuttle
	J	U+E00D	Nassau Street Local
	Jd	U+E00E	Nassau Street Express
	JFK	U+E00F	JFK Express
	K	U+E010	Eighth Avenue Local
	L	U+E011	14th Street–Canarsie Local
	M.brown	U+E012	Nassau Street Local (1979–2010)
	Md.brown	U+E013	Nassau Street Express (2001–2010)
	N	U+E014	Broadway Express
	P	U+E015	Unassigned
	Q	U+E016	Second Avenue Express
	Q.orange	U+E017	Sixth Avenue Express (1988–2001)
	R	U+E018	Broadway Local
	Rd.brown	U+E019	Nassau Street Express (1985–1987)
	S6	U+E01A	Sixth Avenue Shuttle
	SB	U+E01B	63rd Street Shuttle
	SF.silver	U+E01C	Franklin Avenue Shuttle
	T.black	U+E01D	Unassigned
	U	U+E01E	Unassigned
	V	U+E01F	Sixth Avenue Local

Bullet	Key	Code point	Description
	Wd	U+E020	Broadway Express
	X	U+E021	Placeholder for proposed services
	Y	U+E022	Unassigned
	Z	U+E023	Nassau Street Express

4.6 NYC Subway Bullets Retired (set=retired)

Bullet	Key	Code point	Description
	1.1967–1979	U+E000	Broadway–Seventh Avenue Local (1967–1979)
	2.1967–1979	U+E001	Seventh Avenue Express (1967–1979)
	3.1967–1979	U+E002	Seventh Avenue Express (1967–1979)
	4.1967–1979	U+E003	Lexington Avenue Express (1967–1979)
	4d.helv	U+E004	Lexington Avenue Express
	4d.std	U+E005	Lexington Avenue Express
	5.1967–1979	U+E006	Lexington Avenue Express (1967–1979)
	5d.helv	U+E007	Lexington Avenue Express
	5d.std	U+E008	Lexington Avenue Express
	6.1967–1979	U+E009	Lexington Avenue Local (1967–1979)
	7.1967–1979	U+E00A	Flushing Local (1967–1979)
	8.1967–1979	U+E00B	Third Avenue Local (1967–1979)
	8.turquoise	U+E00C	Third Avenue Local
	9.helv	U+E00D	Broadway–Seventh Avenue Local
	9.std	U+E00E	Broadway–Seventh Avenue Local
	A.1967–1979	U+E00F	Eighth Avenue Express (1967–1979)
	A.1967–1987	U+E010	Eighth Avenue Express (1967–1987)
	A.orange	U+E011	Eighth Avenue Express via Sixth Avenue
	Ad.1979–1987	U+E012	Fulton Street Express (1979–1987)
	AA.1967–1979	U+E013	Eighth Avenue Local (1967–1979)
	AA.1979–1985	U+E014	Eighth Avenue Local (1979–1985)
	B.1967–1979	U+E015	Sixth Avenue Express (1967–1979)
	B.yellow	U+E016	Broadway Express (1986–1988)

Bullet	Key	Code point	Description
	Bd.helv	U+E017	Sixth Avenue Express
	Bd.std	U+E018	Sixth Avenue Express
	Bd.yellow	U+E019	Broadway Express (1986–1988)
	Bs	U+E01A	West End Local via Broadway (1986–1988)
	Bsd	U+E01B	West End Express via Broadway (1986–1988)
	C.1985–1987	U+E01C	Eighth Avenue Local (1985–1987)
	Cd	U+E01D	Eighth Avenue Express
	Cd.pre-1987	U+E01E	Eighth Avenue Express (pre-1987)
	CC.1967–1979	U+E01F	Eighth Avenue Local (1967–1979)
	CC.1979–1985	U+E020	Eighth Avenue Local (1979–1985)
	CCd	U+E021	Eighth Avenue Express
	D.1967–1979	U+E022	Sixth Avenue Express (1967–1979)
	D.yellow	U+E023	Sixth Avenue Express, Bronx–34th Street (1986–1988)
	Dd.helv	U+E024	Sixth Avenue Express
	Dd.std	U+E025	Sixth Avenue Express
	Ds	U+E026	Brighton Local via Broadway (1986–1988)
	E.1967–1979	U+E027	Eighth Avenue Local (1967–1979)
	E.1979–1987	U+E028	Eighth Avenue Local (1979–1987)
	EE.1967–1976	U+E029	Broadway Local (1967–1976)
	F.1967–1979	U+E02A	Sixth Avenue Local (1967–1979)
	GG.1967–1979	U+E02B	Brooklyn–Queens Crosstown Local (1967–1979)
	GG.1979–1985	U+E02C	Brooklyn–Queens Crosstown Local (1979–1985)
	H.1985–1987	U+E02D	Rockaway Park Shuttle (1985–1987)
	H.helv	U+E02E	Rockaway Park Shuttle
	H.std	U+E02F	Rockaway Park Shuttle
	HH.1967–1972	U+E030	Court Street Shuttle
	J.1973–1979	U+E031	Nassau Street Local
	Jd.helv	U+E032	Nassau Street Express
	Jd.std	U+E033	Nassau Street Express
	JFK	U+E034	JFK Express
	JJ.1967–1973	U+E035	Nassau Street Local (1967–1973)
	K.1973–1976	U+E036	Jamaica–Sixth Avenue Local (1973–1976)

Bullet	Key	Code point	Description
	K.1985–1987	U+E037	Eighth Avenue Local (1985–1987)
	K.helv	U+E038	Eighth Avenue Local
	K.std	U+E039	Eighth Avenue Local
	KK.1967–1973	U+E03A	Jamaica–Sixth Avenue Local (1967–1973)
	LL.1967–1979	U+E03B	14th Street–Canarsie Local (1967–1979)
	LL.1979–1985	U+E03C	14th Street–Canarsie Local (1979–1985)
	M.1967–1979	U+E03D	Sixth Avenue Local (1967–1979)
	M.brown.helv	U+E03E	Nassau Street Local (1979–2010)
	M.brown.std	U+E03F	Nassau Street Local (1979–2010)
	Md.brown.helv	U+E040	Nassau Street Express (2001–2010)
	Md.brown.std	U+E041	Nassau Street Express (2001–2010)
	MJ.1967–1979	U+E042	Myrtle Avenue Local (1967–1979)
	N.1967–1979	U+E043	Broadway Express (1967–1979)
	Nd.helv	U+E044	Broadway Express
	Nd.std	U+E045	Broadway Express
	NX.1967–1968	U+E046	Sea Beach–Brighton Express (1967–1968)
	Q.orange.helv	U+E047	Sixth Avenue Express (1988–2001)
	Q.orange.std	U+E048	Sixth Avenue Express (1988–2001)
	Qd	U+E049	Second Avenue Express
	QB.1967–1979	U+E04A	Broadway–Brighton Local (1967–1979)
	QBd.1979–1985	U+E04B	Broadway–Brighton Express (1979–1985)
	QJ.1967–1976	U+E04C	Nassau Street–Brighton Local (1967–1976)
	Rd	U+E04D	Broadway Express
	RJ.1967–1968	U+E04E	Jamaica–Fourth Avenue Local (1967–1968)
	RR.1967–1979	U+E04F	Broadway–Fourth Avenue Local (1967–1979)
	RR.1979–1985	U+E050	Broadway–Fourth Avenue Local (1979–1985)
	RRd.yellow	U+E051	Broadway–Fourth Avenue Express
	S.orange	U+E052	63rd Street Shuttle
	S.yellow	U+E053	Lefferts Boulevard Shuttle
	S6	U+E054	Sixth Avenue Shuttle
	SB	U+E055	63rd Street Shuttle
	SS.1967–1968.black	U+E056	145th Street Shuttle (1967–1968)

Bullet	Key	Code point	Description
	SS.1967–1968.blue	U+E057	South Ferry Shuttle (1967–1968)
	SS.1967–1968.orange	U+E058	Culver Shuttle (1967–1968)
	SS.1967–1968.turquoise	U+E059	42nd Street Shuttle (1967–1968)
	SS.1967–1968.yellow	U+E05A	Franklin Avenue Shuttle (1967–1968)
	SS.1967–1979	U+E05B	Shuttle (1967–1979)
	Special.1967–1979	U+E05C	Special service (1967–1979)
	TT.1967	U+E05D	West End Local (1967)
	V.helv	U+E05E	Sixth Avenue Local
	V.std	U+E05F	Sixth Avenue Local
	Wd.helv	U+E060	Broadway Express
	Wd.std	U+E061	Broadway Express
	Zd	U+E062	Nassau Street Express

4.7 NYC Subway Bullets Legacy (set=legacy)

Bullet	Key	Code point	Description
	9.red	U+E000	Broadway–Seventh Avenue Local
	Q.alt	U+E001	Second Avenue Express
	R.alt	U+E002	Broadway Local
	SIR.2016–2017	U+E003	Staten Island Railway (2016–2017)

4.8 NYC Subway Bullets All (set=all)

Bullet	Key	Code point	Description
	–	U+E000	Blank
	–d	U+E001	Blank
	0	U+E002	42nd Street Shuttle
	0d	U+E003	Unassigned
	1	U+E004	Broadway–Seventh Avenue Local
	1.helv	U+E005	Broadway–Seventh Avenue Local

Bullet	Key	Code point	Description
	1.nycta	U+E006	Broadway–Seventh Avenue Local
	1.retired.1967–1979	U+E007	Broadway–Seventh Avenue Local (1967–1979)
	1d	U+E008	Broadway–Seventh Avenue Express
	2	U+E009	Seventh Avenue Express
	2.helv	U+E00A	Seventh Avenue Express
	2.nycta	U+E00B	Seventh Avenue Express
	2.retired.1967–1979	U+E00C	Seventh Avenue Express (1967–1979)
	2d	U+E00D	Seventh Avenue Express
	3	U+E00E	Seventh Avenue Express
	3.helv	U+E00F	Seventh Avenue Express
	3.nycta	U+E010	Seventh Avenue Express
	3.retired.1967–1979	U+E011	Seventh Avenue Express (1967–1979)
	3d	U+E012	Seventh Avenue Express
	4	U+E013	Lexington Avenue Express
	4.helv	U+E014	Lexington Avenue Express
	4.nycta	U+E015	Lexington Avenue Express
	4.retired.1967–1979	U+E016	Lexington Avenue Express (1967–1979)
	4d	U+E017	Lexington Avenue Express
	4d.helv	U+E018	Lexington Avenue Express
	4d.nycta	U+E019	Lexington Avenue Express
	5	U+E01A	Lexington Avenue Express
	5.helv	U+E01B	Lexington Avenue Express
	5.nycta	U+E01C	Lexington Avenue Express
	5.retired.1967–1979	U+E01D	Lexington Avenue Express (1967–1979)
	5d	U+E01E	Lexington Avenue Express
	5d.helv	U+E01F	Lexington Avenue Express
	5d.nycta	U+E020	Lexington Avenue Express
	6	U+E021	Lexington Avenue Local
	6.helv	U+E022	Lexington Avenue Local
	6.nycta	U+E023	Lexington Avenue Local
	6.retired.1967–1979	U+E024	Lexington Avenue Local (1967–1979)
	6d	U+E025	Pelham Express

Bullet	Key	Code point	Description
	6d.helv	U+E026	Pelham Express
	6d.nycta	U+E027	Pelham Express
	7	U+E028	Flushing Local
	7.helv	U+E029	Flushing Local
	7.nycta	U+E02A	Flushing Local
	7.retired.1967–1979	U+E02B	Flushing Local (1967–1979)
	7d	U+E02C	Flushing Express
	7d.helv	U+E02D	Flushing Express
	7d.nycta	U+E02E	Flushing Express
	8	U+E02F	Unassigned
	8.retired.1967–1979	U+E030	Third Avenue Local (1967–1979)
	8.turquoise	U+E031	Third Avenue Local
	8d	U+E032	Unassigned
	8d.turquoise	U+E033	Third Avenue Express
	9	U+E034	Broadway–Seventh Avenue Local
	9.helv	U+E035	Broadway–Seventh Avenue Local
	9.legacy.red	U+E036	Broadway–Seventh Avenue Local
	9.nycta	U+E037	Broadway–Seventh Avenue Local
	9d	U+E038	Broadway–Seventh Avenue Express
	10	U+E039	Unassigned
	11	U+E03A	Unassigned
	12	U+E03B	Unassigned
	13	U+E03C	Unassigned
	A	U+E03D	Eighth Avenue Express
	A.gray	U+E03E	Rockaway Park Shuttle
	A.helv	U+E03F	Eighth Avenue Express
	A.helv.gray	U+E040	Rockaway Park Shuttle
	A.helv.orange	U+E041	Eighth Avenue Express via Sixth Avenue
	A.nycta	U+E042	Eighth Avenue Express
	A.nycta.sixth	U+E043	Eighth Avenue Express via Sixth Avenue
	A.retired.1967–1979	U+E044	Eighth Avenue Express (1967–1979)
	A.retired.1967–1987	U+E045	Eighth Avenue Express (1967–1987)

Bullet	Key	Code point	Description
	Ad	U+E046	Eighth Avenue Express
	Ad.gray	U+E047	Rockaway Park Shuttle Express
	Ad.helv	U+E048	Eighth Avenue Express
	Ad.retired.1979–1987	U+E049	Fulton Street Express (1979–1987)
	AA.retired.1967–1979	U+E04A	Eighth Avenue Local (1967–1979)
	AA.retired.1979–1985	U+E04B	Eighth Avenue Local (1979–1985)
	B	U+E04C	Sixth Avenue Express
	B.helv	U+E04D	Sixth Avenue Express
	B.helv.yellow	U+E04E	Broadway Express (1986–1988)
	B.nycta	U+E04F	Sixth Avenue Express
	B.nycta.broadway	U+E050	Broadway Express
	B.retired.1967–1979	U+E051	Sixth Avenue Express (1967–1979)
	Bd	U+E052	Sixth Avenue Express
	Bd.helv	U+E053	Sixth Avenue Express
	Bd.helv.yellow	U+E054	Broadway Express (1986–1988)
	Bd.nycta	U+E055	Sixth Avenue Express
	Bd.nycta.broadway	U+E056	Broadway Express
	Bs	U+E057	West End Local via Broadway (1986–1988)
	Bsd	U+E058	West End Express via Broadway (1986–1988)
	C	U+E059	Eighth Avenue Local
	C.helv	U+E05A	Eighth Avenue Local
	C.nycta	U+E05B	Eighth Avenue Local
	C.retired.1985–1987	U+E05C	Eighth Avenue Local (1985–1987)
	Cd	U+E05D	Eighth Avenue Express
	Cd.helv	U+E05E	Eighth Avenue Express
	Cd.nycta	U+E05F	Eighth Avenue Express
	Cd.retired.pre-1987	U+E060	Eighth Avenue Express (pre-1987)
	CC.retired.1967–1979	U+E061	Eighth Avenue Local (1967–1979)
	CC.retired.1979–1985	U+E062	Eighth Avenue Local (1979–1985)
	CCd.retired	U+E063	Eighth Avenue Express
	D	U+E064	Sixth Avenue Express
	D.helv	U+E065	Sixth Avenue Express

Bullet	Key	Code point	Description
	D.helv.yellow	U+E066	Sixth Avenue Express, Bronx–34th Street (1986–1988)
	D.nycta	U+E067	Sixth Avenue Express
	D.nycta.broadway	U+E068	Broadway Express
	D.retired.1967–1979	U+E069	Sixth Avenue Express (1967–1979)
	Dd	U+E06A	Sixth Avenue Express
	Dd.helv	U+E06B	Sixth Avenue Express
	Ds	U+E06C	Brighton Local via Broadway (1986–1988)
	Dsd	U+E06D	Brighton Express via Broadway (1986–1988)
	E	U+E06E	Eighth Avenue Local
	E.helv	U+E06F	Eighth Avenue Local
	E.nycta	U+E070	Eighth Avenue Local
	E.retired.1967–1979	U+E071	Eighth Avenue Local (1967–1979)
	E.retired.1979–1987	U+E072	Eighth Avenue Local (1979–1987)
	Ed	U+E073	Eighth Avenue Express
	EE.retired.1967–1976	U+E074	Broadway Local (1967–1976)
	F	U+E075	Sixth Avenue Local
	F.helv	U+E076	Sixth Avenue Local
	F.nycta	U+E077	Sixth Avenue Local
	F.retired.1967–1979	U+E078	Sixth Avenue Local (1967–1979)
	Fd	U+E079	Culver Express
	Fd.helv	U+E07A	Culver Express
	Fd.nycta	U+E07B	Culver Express
	G	U+E07C	Brooklyn–Queens Crosstown Local
	G.helv	U+E07D	Brooklyn–Queens Crosstown Local
	G.nycta	U+E07E	Brooklyn–Queens Crosstown Local
	Gd	U+E07F	Brooklyn–Queens Crosstown Express
	GG.retired.1967–1979	U+E080	Brooklyn–Queens Crosstown Local (1967–1979)
	GG.retired.1979–1985	U+E081	Brooklyn–Queens Crosstown Local (1979–1985)
	H	U+E082	Rockaway Park Shuttle
	H.helv	U+E083	Rockaway Park Shuttle
	H.nycta	U+E084	Rockaway Park Shuttle
	H.retired.1985–1987	U+E085	Rockaway Park Shuttle (1985–1987)

Bullet	Key	Code point	Description
	Hd	U+E086	Rockaway Park Shuttle Express
	HH.retired.1967–1972	U+E087	Court Street Shuttle
	I	U+E088	Unassigned
	Id	U+E089	Unassigned
	J	U+E08A	Nassau Street Local
	J.helv	U+E08B	Nassau Street Local
	J.nycta	U+E08C	Nassau Street Local
	J.retired.1973–1979	U+E08D	Nassau Street Local
	Jd	U+E08E	Nassau Street Express
	Jd.helv	U+E08F	Nassau Street Express
	Jd.nycta	U+E090	Nassau Street Express
	JFK.r32	U+E091	JFK Express
	JJ.retired.1967–1973	U+E092	Nassau Street Local (1967–1973)
	K	U+E093	Eighth Avenue Local
	K.helv	U+E094	Eighth Avenue Local
	K.helv.gray	U+E095	Canarsie skip-stop (proposed)
	K.nycta	U+E096	Eighth Avenue Local
	K.nycta.canarsie	U+E097	Canarsie skip-stop (proposed)
	K.retired.1973–1976	U+E098	Jamaica–Sixth Avenue Local (1973–1976)
	K.retired.1985–1987	U+E099	Eighth Avenue Local (1985–1987)
	Kd	U+E09A	Eighth Avenue Express
	KK.retired.1967–1973	U+E09B	Jamaica–Sixth Avenue Local (1967–1973)
	L	U+E09C	14th Street–Canarsie Local
	L.helv	U+E09D	14th Street–Canarsie Local
	L.nycta	U+E09E	14th Street–Canarsie Local
	Ld	U+E09F	14th Street–Canarsie Express
	LL.retired.1967–1979	U+E0A0	14th Street–Canarsie Local (1967–1979)
	LL.retired.1979–1985	U+E0A1	14th Street–Canarsie Local (1979–1985)
	M	U+E0A2	Sixth Avenue Local
	M.brown	U+E0A3	Nassau Street Local (1979–2010)
	M.helv	U+E0A4	Sixth Avenue Local
	M.helv.brown	U+E0A5	Nassau Street Local (1979–2010)

Bullet	Key	Code point	Description
	M.nycta	U+E0A6	Sixth Avenue Local
	M.nycta.nassau	U+E0A7	Nassau Street Local
	M.retired.1967–1979	U+E0A8	Sixth Avenue Local (1967–1979)
	Md	U+E0A9	Sixth Avenue Express
	Md.brown	U+E0AA	Nassau Street Express (2001–2010)
	Md.helv.brown	U+E0AB	Nassau Street Express (2001–2010)
	Md.nycta.nassau	U+E0AC	Nassau Street Express
	MJ.retired.1967–1979	U+E0AD	Myrtle Avenue Local (1967–1979)
	N	U+E0AE	Broadway Express
	N.helv	U+E0AF	Broadway Express
	N.nycta	U+E0B0	Broadway Express
	N.retired.1967–1979	U+E0B1	Broadway Express (1967–1979)
	Nd	U+E0B2	Broadway Express
	Nd.helv	U+E0B3	Broadway Express
	Nd.nycta	U+E0B4	Broadway Express
	NX.retired.1967–1968	U+E0B5	Sea Beach–Brighton Express (1967–1968)
	O	U+E0B6	Unassigned
	Od	U+E0B7	Unassigned
	P	U+E0B8	Unassigned
	Pd	U+E0B9	Unassigned
	Q	U+E0BA	Second Avenue Express
	Q.helv	U+E0BB	Second Avenue Express
	Q.legacy.alt	U+E0BC	Second Avenue Express
	Q.nycta	U+E0BD	Second Avenue Express
	Q.nycta.sixth	U+E0BE	Sixth Avenue Express (1988–2001)
	Q.orange	U+E0BF	Sixth Avenue Express (1988–2001)
	Q.retired.orange.helv	U+E0C0	Sixth Avenue Express (1988–2001)
	Qd	U+E0C1	Second Avenue Express
	Qd.helv	U+E0C2	Second Avenue Express
	Qd.nycta	U+E0C3	Second Avenue Express
	Qd.orange	U+E0C4	Sixth Avenue Express (1988–2001)
	QB.retired.1967–1979	U+E0C5	Broadway–Brighton Local (1967–1979)

Bullet	Key	Code point	Description
	QBd.retired.1979–1985	U+E0C6	Broadway–Brighton Express (1979–1985)
	QJ.retired.1967–1976	U+E0C7	Nassau Street–Brighton Local (1967–1976)
	R	U+E0C8	Broadway Local
	R.brown	U+E0C9	Nassau Street Local (1985–1987)
	R.helv	U+E0CA	Broadway Local
	R.legacy.alt	U+E0CB	Broadway Local
	R.nycta	U+E0CC	Broadway Local
	Rd	U+E0CD	Broadway Express
	Rd.brown	U+E0CE	Nassau Street Express (1985–1987)
	Rd.helv	U+E0CF	Broadway Express
	Rd.nycta	U+E0D0	Broadway Express
	Rd.nycta.nassau	U+E0D1	Nassau Street Express
	RJ.retired.1967–1968	U+E0D2	Jamaica–Fourth Avenue Local (1967–1968)
	RR.retired.1967–1979	U+E0D3	Broadway–Fourth Avenue Local (1967–1979)
	RR.retired.1979–1985	U+E0D4	Broadway–Fourth Avenue Local (1979–1985)
	RRd.retired.yellow	U+E0D5	Broadway–Fourth Avenue Express
	S	U+E0D6	Shuttle
	S.helv	U+E0D7	Shuttle
	S.helv.blue	U+E0D8	Rockaway Park Shuttle
	S.helv.orange	U+E0D9	63rd Street Shuttle
	S.helv.yellow	U+E0DA	Lefferts Boulevard Shuttle
	S.nycta	U+E0DB	Shuttle
	S.nycta.63	U+E0DC	63rd Street Shuttle
	S.nycta.rockaway	U+E0DD	Rockaway Park Shuttle
	S.nycta.sixth	U+E0DE	Sixth Avenue Shuttle
	Sd	U+E0DF	Shuttle Express
	S6	U+E0E0	Sixth Avenue Shuttle
	S6d	U+E0E1	Sixth Avenue Shuttle Express
	SB	U+E0E2	63rd Street Shuttle
	SBd	U+E0E3	63rd Street Shuttle Express
	SF	U+E0E4	Franklin Avenue Shuttle
	SF.helv	U+E0E5	Franklin Avenue Shuttle

Bullet	Key	Code point	Description
	SF.silver	U+E0E6	Franklin Avenue Shuttle
	SFd.silver	U+E0E7	Franklin Avenue Shuttle Express
	SIR	U+E0E8	Staten Island Railway
	SIR.helv	U+E0E9	Staten Island Railway
	SIR.legacy.2016–2017	U+E0EA	Staten Island Railway (2016–2017)
	SIRd	U+E0EB	Staten Island Railway Express
	SR	U+E0EC	Rockaway Park Shuttle
	SR.blue	U+E0ED	Rockaway Park Shuttle
	SR.helv	U+E0EE	Rockaway Park Shuttle
	SRd.blue	U+E0EF	Rockaway Park Shuttle Express
	SS.retired.1967–1968.black	U+E0F0	145th Street Shuttle (1967–1968)
	SS.retired.1967–1968.blue	U+E0F1	South Ferry Shuttle (1967–1968)
	SS.retired.1967–1968.orange	U+E0F2	Culver Shuttle (1967–1968)
	SS.retired.1967–1968.turquoise	U+E0F3	42nd Street Shuttle (1967–1968)
	SS.retired.1967–1968.yellow	U+E0F4	Franklin Avenue Shuttle (1967–1968)
	SS.retired.1967–1979	U+E0F5	Shuttle (1967–1979)
S	Special.retired.1967–1979	U+E0F6	Special service (1967–1979)
	T	U+E0F7	Second Avenue Local (proposed)
	T.black	U+E0F8	Unassigned
	T.helv	U+E0F9	Second Avenue Local (proposed)
	T.nycta	U+E0FA	Second Avenue Local (proposed)
	Td	U+E0FB	Second Avenue Express (proposed)
	Td.black	U+E0FC	Unassigned
	TT.retired.1967	U+E0FD	West End Local (1967)
	U	U+E0FE	Unassigned
	Ud	U+E0FF	Unassigned
	V	U+E100	Sixth Avenue Local
	V.helv	U+E101	Sixth Avenue Local
	V.nycta	U+E102	Sixth Avenue Local
	Vd	U+E103	Sixth Avenue Express
	W	U+E104	Broadway Local
	W.helv	U+E105	Broadway Local

Bullet	Key	Code point	Description
	W.nycta	U+E106	Broadway Local
	Wd	U+E107	Broadway Express
	Wd.helv	U+E108	Broadway Express
	Wd.nycta	U+E109	Broadway Express
	X	U+E10A	Placeholder for proposed services
	Xd	U+E10B	Placeholder for proposed services
	Y	U+E10C	Unassigned
	Yd	U+E10D	Unassigned
	Z	U+E10E	Nassau Street Express
	Z.helv	U+E10F	Nassau Street Express
	Z.nycta	U+E110	Nassau Street Express
	Zd	U+E111	Nassau Street Express
	Zd.helv	U+E112	Nassau Street Express

5 nycbullets implementation

```

1 <*package>
2 <@@=nycbullets>

3 \NeedsTeXFormat{LaTeX2e}[2022-06-01]
4 \ProvidesExplPackage{nycbullets}{2026-10-03}{0.1}
5   {New York City Subway route bullets}

```

5.1 Messages

```

6 \msg_new:nnn { nycbullets } { pdftex }
7   {
8     The~nycbullets~package~needs~OpenType~fonts:~
9     use~LuaLaTeX~(or~XeLaTeX~for~monochrome~bullets).
10  }
11 \msg_new:nnn { nycbullets } { xetex }
12   { XeTeX~cannot~render~color~fonts:~the~bullets~will~be~monochrome. }
13 \msg_new:nnnn { nycbullets } { unknown-key }
14   { Unknown~bullet~key~`#1'~in~set~`#2'. }
15   { See~the~nycbullets~documentation~for~the~keys~in~each~set. }
16 \msg_new:nnn { nycbullets } { unknown-option }
17   { Unknown~option~`#1'. }
18 \msg_new:nnnn { nycbullets } { unknown-set }
19   { Unknown~bullet~set~`#1'. }
20   { The~known~sets~are:~#2. }

```

5.2 Engine check

```

21 \sys_if_engine_pdftex:T
22 {
23     \msg_critical:nn { nycbullets } { pdftex }
24 }
25 \sys_if_engine_xetex:T
26 { \msg_warning:nn { nycbullets } { xetex } }
27 \RequirePackage { fontspec }

```

5.3 Variables

```

28 \tl_new:N \l__nycbullets_set_tl
29 \fp_new:N \l__nycbullets_scale_fp
30 \dim_new:N \l__nycbullets_raise_dim
31 \bool_new:N \l__nycbullets_color_bool
32 \tl_new:N \l__nycbullets_renderer_tl
33 \tl_new:N \l__nycbullets_id_tl
34 \tl_new:N \l__nycbullets_cp_tl
35 \box_new:N \l__nycbullets_box

```

Set id → font file; any accepted set name → set id; the list of set ids (for messages).

```

36 \prop_new:N \g__nycbullets_sets_prop
37 \prop_new:N \g__nycbullets_alias_prop
38 \seq_new:N \g__nycbullets_names_seq

```

(End of definition for `\g__nycbullets_sets_prop`, `\g__nycbullets_alias_prop`, and `\g__nycbullets_names_seq`.)

5.4 Declaring sets and keys

These are used by the generated files `nycbullets-sets.def` and `nycbullets-⟨set⟩.def`.

```
\nycbullets_declare_set:nnn \nycbullets_declare_set:nnn {⟨id⟩} {⟨name⟩} {⟨font file⟩}
```

```

39 \cs_new_protected:Npn \nycbullets_declare_set:nnn #1#2#3
40 {
41     \prop_gput:Nnn \g__nycbullets_sets_prop {#1} {#3}
42     \prop_gput:Nnn \g__nycbullets_alias_prop {#1} {#1}
43     \prop_gput:Nen \g__nycbullets_alias_prop { \str_lowercase:n {#2} } {#1}
44     \seq_gput_right:Nn \g__nycbullets_names_seq {#1}
45     \prop_new:c { g__nycbullets_keys_#1_prop }
46     \bool_new:c { g__nycbullets_loaded_#1_bool }
47 }

```

(End of definition for `\nycbullets_declare_set:nnn`. This function is documented on page ??.)

```
\nycbullets_declare_keys:nn \nycbullets_declare_keys:nn {⟨id⟩} { {⟨key⟩} {⟨hex code point⟩} ... }
```

```

48 \cs_new_protected:Npn \nycbullets_declare_keys:nn #1#2
49 {
50     \bool_set_false:N \l__nycbullets_pending_bool
51     \tl_map_tokens:nn {#2} { \__nycbullets_declare_key:nn {#1} }
52 }

```

`\tl_map_tokens:nn` hands over one braced item at a time, so keys and code points alternate: we remember each key until its code point arrives.

```

53 \tl_new:N \l__nycbullets_pending_key_tl
54 \bool_new:N \l__nycbullets_pending_bool
55 \cs_new_protected:Npn \__nycbullets_declare_key:nn #1#2
56 {
57   \bool_if:NTF \l__nycbullets_pending_bool
58   {
59     \prop_gput:cVn { g__nycbullets_keys_#1_prop } \l__nycbullets_pending_key_tl {
60       \bool_set_false:N \l__nycbullets_pending_bool
61     }
62   }
63   \tl_set:Nc \l__nycbullets_pending_key_tl { \tl_to_str:n {#2} }
64   \bool_set_true:N \l__nycbullets_pending_bool
65 }
66 }

```

(End of definition for `\nycbullets_declare_keys:nn`. This function is documented on page ??.)

```

67 \file_input:n { nycbullets-sets.def }

```

5.5 Loading a set

`\__nycbullets_resolve_set:n` Set `\l__nycbullets_id_tl` to the id of the set named `#1`, or raise an error and fall back to the first set.

```

68 \cs_new_protected:Npn \__nycbullets_resolve_set:n #1
69 {
70   \prop_get:NcNF \g__nycbullets_alias_prop { \str_lowercase:n {#1} } \l__nycbullets
71   {
72     \msg_error:nnne { nycbullets } { unknown-set } {#1}
73     { \seq_use:Nn \g__nycbullets_names_seq { ,~ } }
74     \tl_set:Nc \l__nycbullets_id_tl { \seq_item:Nn \g__nycbullets_names_seq { 1 } }
75   }
76 }

```

(End of definition for `\__nycbullets_resolve_set:n`.)

`\__nycbullets_load:V` On first use of a set, read its key table. This may happen inside a group, so everything it defines is global. We make sure `%` is a comment character, which it isn't in (for instance) `\DocInput`, and that line ends are ignored, so the file can't end the paragraph.

```

77 \cs_new_protected:Npn \__nycbullets_load:n #1
78 {
79   \bool_if:cF { g__nycbullets_loaded_#1_bool }
80   {
81     \group_begin:
82       \char_set_catcode_comment:N \%
83       \char_set_catcode_ignore:n { 13 }
84       \file_input:n { nycbullets-#1.def }
85     \group_end:
86     \bool_gset_true:c { g__nycbullets_loaded_#1_bool }
87   }
88 }
89 \cs_generate_variant:Nn \__nycbullets_load:n { V }

```

(End of definition for `\__nycbullets_load:V`.)

`\__nycbullets_declare_font:nn` Declare the font family `\@@_font_<id>:.` With LuaT_EX the `colr` feature turns on the color layers.

```

90 \cs_new_protected:Npn \__nycbullets_declare_font:nn #1#2
91 {
92   \clist_clear:N \__nycbullets_features_clist
93   \sys_if_engine luatex:T
94   {
95     \str_if_eq:VnT \__nycbullets_renderer_tl { harfbuzz }
96     { \clist_put_right:Nn \__nycbullets_features_clist { Renderer = HarfBuzz } }
97     \bool_if:NTF \__nycbullets_color_bool
98     { \clist_put_right:Nn \__nycbullets_features_clist { RawFeature = {+colr} } }
99     { \clist_put_right:Nn \__nycbullets_features_clist { RawFeature = {-
colr} } } }
100   }
101   \use:e
102   {
103     \exp_not:N \newfontfamily
104     \exp_not:c { __nycbullets_font_#1: }
105     { \exp_not:n {#2} }
106     [ \exp_not:V \__nycbullets_features_clist ]
107   }
108 }
109 \clist_new:N \__nycbullets_features_clist

```

(End of definition for `\__nycbullets_declare_font:nn`.)

The font families are declared at the start of the document, after any `\nycbulletsetup` in the preamble. (`\newfontfamily` defines its command locally, so it cannot wait for the first bullet, which is typeset in a group.) Declaring a family does not load the font.

```

110 \cs_new_protected:Npn \__nycbullets_declare_fonts:
111 {
112   \prop_map_inline:Nn \g__nycbullets_sets_prop
113   { \__nycbullets_declare_font:nn {##1} {##2} }
114 }
115 \hook_gput_code:nnn { begindocument } { nycbullets } { \__nycbullets_declare_fonts: }

```

5.6 Options

```

116 \keys_define:nn { nycbullets }
117 {
118   set      .tl_set:N    = \__nycbullets_set_tl ,
119   set      .initial:n   = std ,
120   scale    .fp_set:N    = \__nycbullets_scale_fp ,
121   scale    .initial:n   = 1 ,
122   raise    .dim_set:N   = \__nycbullets_raise_dim ,
123   raise    .initial:n   = 0pt ,
124   color    .bool_set:N  = \__nycbullets_color_bool ,
125   color    .initial:n   = true ,
126   color    .default:n   = true ,
127   renderer .choices:nn = { node , harfbuzz }
128   { \tl_set:Nc \__nycbullets_renderer_tl { \l_keys_choice_tl } } ,
129   renderer .initial:n   = node ,

```

```

130     unknown .code:n      =
131     {
132         \tl_if_blank:nTF {#1}
133         { \tl_set:Nc \l__nycbullets_set_tl { \l_keys_key_str } }
134         {
135             \msg_error:nne { nycbullets } { unknown-option }
136             { \l_keys_key_str }
137         }
138     } ,
139 }
140 \ProcessKeyOptions [ nycbullets ]

```

5.7 User commands

`\nycbullets_codepoint:nn` The (hexadecimal) code point of key #2 in set #1 (an id or alias), or nothing. Expandable; the set must have been used already.

```

141 \cs_new:Npn \nycbullets_codepoint:nn #1#2
142 {
143     \prop_item:cn
144     {
145         g__nycbullets_keys_
146         \exp_args:NNe \prop_item:Nn \g__nycbullets_alias_prop { \str_lowercase:n {#1} }
147         _prop
148     }
149     {#2}
150 }

```

(End of definition for \nycbullets_codepoint:nn. This function is documented on page ??.)

`\nycbullets_bullet:n` Typeset one bullet with the current options.

```

151 \cs_new_protected:Npn \nycbullets_bullet:n #1
152 {
153     \__nycbullets_resolve_set:V \l__nycbullets_set_tl
154     \__nycbullets_load:V \l__nycbullets_id_tl
155     \__nycbullets_bullet:Vn \l__nycbullets_id_tl {#1}
156 }

```

The set id is passed as an argument here so that it can be part of control sequence names: in `\ExplSyntaxOn`, `\l_@@_id_tl _prop` would be read as a single name.

```

157 \cs_new_protected:Npn \__nycbullets_bullet:nn #1#2
158 {
159     \prop_get:cnNTF { g__nycbullets_keys_#1_prop } {#2} \l__nycbullets_cp_tl
160     {
161         \mode_leave_vertical:
162         \hbox_set:Nn \l__nycbullets_box
163         {
164             \__nycbullets_select_font:n {#1}
165             \tex_char:D " \l__nycbullets_cp_tl \scan_stop:
166         }
167         \box_move_up:nn { \l__nycbullets_raise_dim } { \box_use_drop:N \l__nycbullets_box }
168     }
169     { \msg_error:nnnn { nycbullets } { unknown-key } {#2} {#1} }

```

```

170 }
171 \cs_generate_variant:Nn \__nycbullets_bullet:nn { V }
172 \cs_generate_variant:Nn \__nycbullets_resolve_set:n { V }

```

(End of definition for `\nycbullets_bullet:n`. This function is documented on page ??.)

`\__nycbullets_select_font:n` Switch to the font of set #1 at the requested scale.

```

173 \cs_new_protected:Npn \__nycbullets_select_font:n #1
174 {
175   \use:c { __nycbullets_font_#1: }
176   \fontsize
177     { \fp_eval:n { \l__nycbullets_scale_fp * \use:c { f@size } } pt }
178     { \use:c { f@baselineskip } }
179   \selectfont
180 }
181 \cs_generate_variant:Nn \__nycbullets_select_font:n { V }

```

(End of definition for `\__nycbullets_select_font:n`.)

`\nycbullet`
`\nycbulletsetup`
`\nycbulletfont`

```

182 \NewDocumentCommand \nycbullet { o m }
183 {
184   \group_begin:
185     \IfValueT {#1} { \keys_set:nn { nycbullets } {#1} }
186     \nycbullets_bullet:n {#2}
187   \group_end:
188 }
189 \NewDocumentCommand \nycbulletsetup { m }
190 { \keys_set:nn { nycbullets } {#1} }
191 \NewDocumentCommand \nycbulletfont { o }
192 {
193   \IfValueT {#1} { \tl_set:Nn \l__nycbullets_set_tl {#1} }
194   \__nycbullets_resolve_set:V \l__nycbullets_set_tl
195   \__nycbullets_load:V \l__nycbullets_id_tl
196   \__nycbullets_select_font:V \l__nycbullets_id_tl
197 }

```

(End of definition for `\nycbullet`, `\nycbulletsetup`, and `\nycbulletfont`. These functions are documented on page 1.)

```

198 \endpackage

```

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